

MO DELIVERABLE · TECHNICAL SOLUTION

PREPARED FOR MOUNTAIN PROPERTY

# Technical Solution Package

Architecture, data flow, AI pipeline, security and hosting — the engineering plan behind Phase 1 MVP.

Issued as the first milestone deliverable under the signed Statement of Works.

PREPARED BY

**Metatree AI Lab**

AI consulting division · JR Academy

MILESTONE M0 · AUD \$5,000

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CONFIDENTIAL · MOUNTAIN PROPERTY INTERNAL USE ONLY

PLATFORM AT A GLANCE

# Five coordinated components — one operating system for land feasibility.

The Phase 1 platform is built as a small number of well-defined services. Each has one responsibility, a clear interface, and can be scaled or replaced independently. Every line of code runs in Mountain Property's cloud account.

|    |                                                                     |                                                                                                                                            |
|----|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 01 | <b>Enterprise Portal</b><br>VITE 6 · REACT 19 · SHADCN/UI           | Day-to-day workspace for Mountain Property users — search sites, generate feasibility reports, manage the land pipeline.                   |
| 02 | <b>Admin Console</b><br>VITE 6 · REACT 19 · SHADCN/UI               | Super-admin surface — users, organisations, audit logs, system config. Only Mountain Property admins.                                      |
| 03 | <b>API Server</b><br>NESTJS 11 · TYPEORM · JWT                      | Unified REST backend at /api/v1 — auth, RBAC, documents, pipeline, report jobs.                                                            |
| 04 | <b>Data &amp; ETL</b><br>PYTHON · PREFECT · DBT · POSTGIS           | Ingests government planning data nationally; normalises Bronze → Silver → Gold.                                                            |
| 05 | <b>AI Pipeline</b><br>MULTI-AGENT CREW · CLAUDE + OPENAI · PGVECTOR | Planner + specialist agents + critic + composer. RAG over Mountain Property's documents + gov data. Multi-model and multi-agent by design. |

## 1 API

SINGLE BACKEND, SINGLE AUTH

## AU only

DATA RESIDENCY GUARANTEED

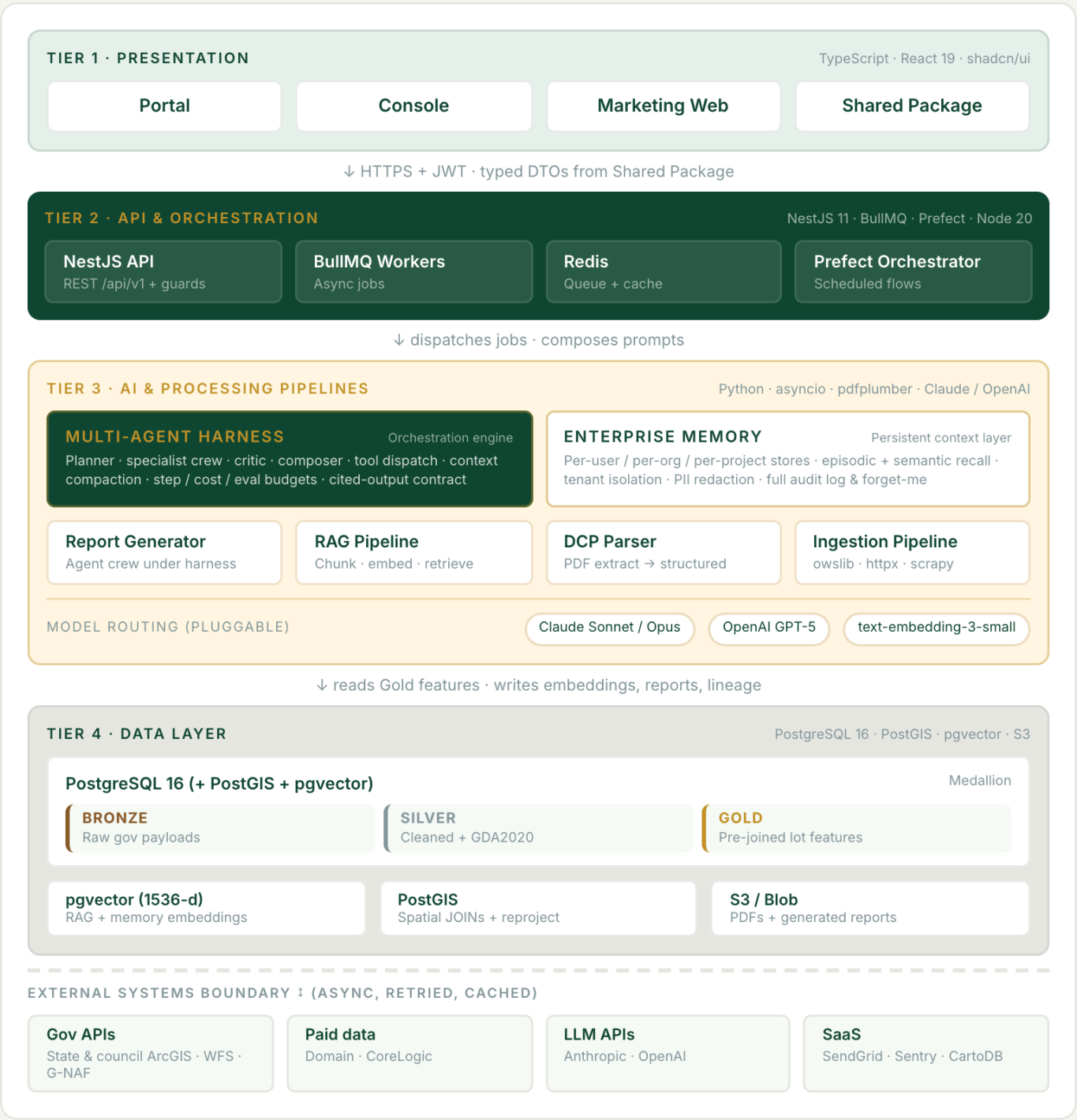
## Zero lock-in

ALL IP TRANSFERS TO YOU

SYSTEM ARCHITECTURE

Four coordinated tiers, a multi-agent harness, and an enterprise-grade memory layer.

Request flow is top-down; ingestion and AI jobs run asynchronously on the right. Tier 3 is centred on two engineered systems Mountain Property owns outright: a **multi-agent harness** that orchestrates planner + specialist + critic + composer agents under strict cost, step, and eval budgets, and an **enterprise memory system** that gives every agent tenant-isolated, audit-logged, long-lived context across projects, users, and workflows.



CLOUD DEPLOYMENT & DATABASE

# Deployed into Mountain Property's own cloud — four zones, one database foundation.

Every service runs inside Mountain Property's AWS or Azure account, AU region. Shown here with AWS service names; Azure equivalents apply one-for-one if that's your preference.

EDGE

Public — DNS, CDN, WAF

CloudFront / Front Door

CDN + TLS termination

Route 53 / Azure DNS

Custom domain

WAF + rate limiting

Basic attack surface

COMPUTE

Private VPC / vNet — stateless services

NestJS API

ECS Fargate / App Service

BullMQ Workers

Async job processors

Prefect agents

ETL + schedule runners

DATA

Isolated subnet — private only

PostgreSQL 16

RDS Multi-AZ / Flexible Server

Redis 7

ElastiCache / Azure Cache

S3 / Blob Storage

Documents, reports, Bronze

SECURITY / OPS

Cross-cutting

Secrets Manager / Key Vault

Credentials & API keys

CloudWatch / Azure Monitor

Logs + metrics + alerts

Automated snapshots + PITR

7-day point-in-time recovery

DATABASE FOUNDATION

Encrypted at rest & in transit · AU region only

| COMPONENT                  | CONFIGURATION                                      | ROLE                                         |
|----------------------------|----------------------------------------------------|----------------------------------------------|
| PostgreSQL 16              | db.m6g.large (prod) · Multi-AZ · encrypted at rest | Primary OLTP + spatial + vector store        |
| PostGIS extension          | GIST / SP-GIST indexes · EPSG:7856                 | Spatial JOINS, reprojection, topology ops    |
| pgvector extension         | ivfflat · 1536-d · cosine similarity               | RAG embeddings — sub-100ms nearest-neighbour |
| Redis 7                    | cache.m6g.large · AOF persistence                  | BullMQ queues + session cache                |
| Object storage (S3 / Blob) | 3 buckets · versioning · SSE-KMS · lifecycle to IA | Raw PDFs, generated reports, Bronze archive  |

AU region

SYDNEY / AU EAST ONLY

Multi-AZ

AUTO-FAILOVER

Encrypted

AT REST + IN TRANSIT

PITR 7d

POINT-IN-TIME RECOVERY

IaC

TERRAFORM — REPRODUCIBLE

NATIONAL DATA ENGINE

The hardest part of the platform — and the reason it works.

Australian planning data is not a single API. It's hundreds of state and council endpoints across six planning regimes, each with its own protocol, projection, schema, and update cadence. Pulling address-in / feasibility-out in under a second requires more than an integration — it requires data engineering, spatial GIS, and data science working together.

14M+

PARCELS NATIONWIDE

500+

COUNCIL DCPs

6

STATE PLANNING REGIMES

30+

HETEROGENEOUS SOURCES

BRONZE · SILVER · GOLD — MEDALLION ARCHITECTURE

BRONZE

Raw gov responses archived untouched in S3 + Postgres

SILVER

Cleaned + reprojected to GDA2020; consistent schema

GOLD

Pre-joined lot feature views — the Portal’s query target

| SOURCE                           | PROTOCOL                      | WHAT WE GET                                               |
|----------------------------------|-------------------------------|-----------------------------------------------------------|
| State planning portals           | ArcGIS REST · WFS · OpenData  | Zoning, FSR, height, heritage, lot size overlays          |
| State hazard layers              | ArcGIS REST                   | Bushfire, flood planning, landslip — multiple agencies    |
| National address + cadastre      | G-NAF · state WFS · Shapefile | Address ↔ lot polygon resolution                          |
| Council DCP PDFs (500+ councils) | Custom crawler + PDF parse    | Setback, height, FSR rules extracted to structured tables |
| Transport network                | GTFS static                   | Stops, routes, transit corridors                          |
| Demographics                     | ABS data cubes (SA1 / SA2)    | Population, dwelling mix, income — for market context     |
| Comparable sales                 | Domain · CoreLogic (paid)     | Recent transactions — yield + ROI inputs                  |
| Aerial imagery + contours        | WMS / tileserver              | Topography, visual context, slope-based feasibility       |

DISCIPLINES BROUGHT TO BEAR

Data Engineering

Prefect ETL, medallion architecture, incremental idempotent ingestion, dbt transforms

Spatial / GIS

PostGIS, GDA2020 / EPSG:7856 reprojection, topology fixes, spatial JOINS

Data Science

Feature engineering, aggregation, risk scoring, evaluation against known outcomes

Ops discipline

Data-quality tests, schema drift alerts, retry + backoff, audit trails



MULTI-AGENT REPORT SYSTEM & HARNESS

# A crew of agents — planner, specialists, critic, composer.

A 20-page institutional feasibility report cannot come from one prompt, or even one agent. Phase 1 ships a **multi-agent system**: a planner scopes the work, several domain-specialist agents run in parallel, an independent critic verifies, and a composer assembles. Every step is wrapped in an **evaluation harness**.

THE CREW — 4 AGENT ROLES

Plan → Specialists (parallel) → Critic → Compose

01

Planner agent

Reads the site brief + Mountain Property template. Scopes sections and writes a typed task brief for each specialist.

02

Specialist agents  
N PARALLEL

One agent per domain, each with a narrow prompt + tool set. Run concurrently, independently retryable.

ZoningAgent

YieldAgent

HazardAgent

DCPComplianceAgent

ComparablesAgent

ROIAgent

03

Critic agent

Independent reviewer — never sees the specialist's chain-of-thought, only its claims + tool outputs. Verifies citations, arithmetic, internal consistency. Fails send the section back to its specialist.

04

Composer agent

Merges verified sections into the final deliverable. Enforces template order, page numbering, cross-references, and render to PDF / DOCX.

Why multi-agent

Narrower context per specialist → higher per-section accuracy.

Parallelism

Specialists run concurrently — end-to-end latency stays flat as sections grow.

Independent critic

Critic never sees the draft's reasoning — catches errors specialist can't see in itself.

| SHARED TOOL CATALOGUE                   |                                                         | Typed · enforced by agent framework |
|-----------------------------------------|---------------------------------------------------------|-------------------------------------|
| site_profile(address)                   | Gold-layer lot features — zoning, FSR, height, overlays |                                     |
| rag_search(query, k)                    | Top-k chunks from Client document corpus (pgvector)     |                                     |
| dcp_rules(lga, controls[])              | Structured DCP rules — setback, height, parking         |                                     |
| yield_calc(fsr, lot_area, efficiency)   | GFA + dwelling-yield estimate                           |                                     |
| roi_calc(revenue, cost, time, discount) | Indicative margin, IRR                                  |                                     |
| comps_search(lot, radius, filters)      | Recent sales — Domain / CoreLogic                       |                                     |
| cite(fact, source_id)                   | Enforced citation — no cite, critic                     |                                     |

EVALUATION HARNESS

The harness is what keeps a multi-agent crew reliable over time. Without it, prompt drift destroys quality silently.

Per-agent eval

Each specialist has its own golden set + rubric, evaluated independently

Crew-level eval

End-to-end runs on Mountain Property benchmark sites, scored against known outcomes

CI eval gate

Every prompt / agent PR runs the harness; regressions block merge

Observability

Per-agent token cost, latency, retry count, tool-call trace to telemetry

A/B across models

Any agent can be routed to Claude or OpenAI — results diffed

## PLATFORM INTEGRATIONS

# Multi-vendor by design — yours to swap whenever.

No component locks Mountain Property into a single LLM, cloud, or SaaS vendor. Every integration sits behind an adapter and can be swapped at configuration level — no platform rewrite required.

## AI MODELS

**Pluggable — never locked to one vendor.**

### Anthropic Claude (Sonnet / Opus)

Primary — long-context reasoning, report generation

### OpenAI GPT-5

Alternative — benchmark + fallback

### OpenAI text-embedding-3-small

Default embeddings (1536-d) for pgvector

### Model router + cost ceilings

Per-feature model config, hard \$/month caps

### Evaluation harness

Prompts benchmarked across models in CI

## CLOUD INFRASTRUCTURE

**In Mountain Property's own account, AU region.**

### AWS (Sydney) or Azure (AU East)

Mountain Property's choice; IaC (Terraform) either way

### PostgreSQL 16 + PostGIS + pgvector

Managed RDS / Azure Flexible Server

### S3 / Blob Storage

Documents, generated reports, Bronze archive

### ElastiCache Redis / Azure Cache

BullMQ queues + session cache

### Secrets Manager / Key Vault

All credentials — nothing in env / source

### CloudFront / Front Door

CDN for Portal and Console static assets

## MANAGED SAAS

**Third-party services the platform integrates with.**

### SendGrid / Resend

Transactional email — invites, resets, report delivery

### Sentry

Error tracking + performance monitoring

### CartoDB map tiles

Free basemap, no API key required

### Google / Microsoft OAuth

Optional SSO for Mountain Property workspace

### GitHub Actions

CI/CD — builds, tests, deploys

### Government planning APIs (national)

Free state + council data — no key, no contract

## Adapter pattern

Every vendor behind a thin interface — swap without touching feature code.

## Cost pass-through

All API / SaaS fees billed directly to Mountain Property after handover.

## AU data residency

Every service configured to keep Mountain Property's data in Australia.

✗ **Self-hosted / on-premise LLM deployment is out of Phase 1 scope.** Phase 1 uses cloud LLM APIs (Claude / OpenAI). If Mountain Property later needs an LLM running on its own servers, that work is quoted separately.

SECURITY, PRIVACY & HOSTING

Built on Mountain Property's cloud. Private by default.

The platform is deployed into Mountain Property's own AWS or Azure account from day one. Metatree AI Lab never holds production data; access is operationally delegated and can be revoked in minutes. Privacy is a first-class design input, not a checkbox.

01

**Data residency**  
All storage, compute, and LLM traffic routed through AU regions. No offshore egress for Mountain Property data.

02

**Tenant isolation**  
Every tenant-scoped table carries organisation\_id; all queries scope through it. Row-level guarantees enforced in the API.

03

**Auth & RBAC**  
JWT access + refresh tokens, bcrypt hashing (≥ 12 rounds), 5-role org model (owner / admin / manager / member / viewer).

04

**Secrets & keys**  
Managed via cloud-native secret store (AWS Secrets Manager / Azure Key Vault). Nothing in source or env files.

05

**Audit trail**  
Every privileged action (user add/remove, role change, doc delete, report export) written to an immutable audit table.

06

**Backup & recovery**  
Automated daily Postgres snapshots + S3 object versioning; tested restore runbook handed over at M4.

07

PRIVACY

**Privacy by design**  
Australian Privacy Principles (APP) alignment. PII minimised, encrypted at rest + in transit, retention policy per data class, export + erasure on request.

08

PRIVACY

**No model training on Client data**  
LLM vendors are contracted in no-retention / no-training mode — Mountain Property's prompts and documents are never used to train foundation models.

| PRIVACY — WHAT IT MEANS CONCRETELY |                                                                                                        | Aligned to Australian Privacy Principles |
|------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------|
| Legal baseline                     | Privacy Act 1988 + Australian Privacy Principles (APP 1 – 13)                                          |                                          |
| Encryption                         | TLS 1.2+ in transit · AES-256 (SSE-KMS) at rest · column-level for PII                                 |                                          |
| Retention                          | Per-class policy — raw gov data cached 30d, user PII retained while account active + 30d               |                                          |
| Subject rights                     | Export + erasure APIs — respond within 30 days per APP 12/13                                           |                                          |
| LLM boundary                       | Anthropic + OpenAI zero-retention flag set · prompts not logged by vendor · embeddings stay in your DB |                                          |



## DELIVERY APPROACH

## Two responsibilities. One clean interface.

The engagement is deliberately low-friction. Metatree AI Lab ships engineering; Mountain Property provides direction, accounts, and domain input. Anything outside these two lists is a written change order.

## METATREE AI LAB DELIVERS

### Engineering, design, AI, infra, handover.

- ◆ Architecture & build — API, Portal, Console, Data Engine, AI Pipeline
- ◆ Integration work against government planning endpoints
- ◆ AI prompt engineering, RAG pipeline, and report templates
- ◆ Cloud infra-as-code (Terraform) for AWS or Azure
- ◆ Source-code, documentation, training, and production cut-over
- ◆ 30-day post-launch hypercare after M4

## MOUNTAIN PROPERTY PROVIDES

### Direction, accounts, and domain input.

- ◆ One product owner — ~2 hours per week (sign-off, priorities, access)
- ◆ 3–5 reference sites with known outcomes (calibration set)
- ◆ Existing feasibility report samples + preferred template structure
- ◆ Cloud account (AWS or Azure, AU region) + commercial LLM API key
- ◆ Access to any internal documents to be ingested into RAG corpus
- ◆ Legal / compliance review sign-off when requested

## WORKING RHYTHM

Weekly 30-min review · stage-end demo + written acceptance · shared Linear/GitHub visibility · response SLA 1 business day during build.

## M0 ACCEPTANCE

# This pack is the M0 deliverable.

Sign-off of this Technical Solution Package unlocks the first milestone invoice of AUD \$5,000 and authorises Metatree AI Lab to begin Stage 1 — Foundation.

### THIS DOCUMENT COVERS

- ✓ Platform overview & apps
- ✓ National data engine design
- ✓ Security & data residency
- ✓ Delivery responsibilities
- ✓ System architecture
- ✓ AI report pipeline
- ✓ Hosting options
- ✓ Working rhythm & SLAs

## ACCEPTANCE OF M0

For Mountain Property

\_\_\_\_\_  
Name · Title · Date

For Metatree AI Lab

\_\_\_\_\_  
Name · Director · Date